

## Literature status: planar and doubling metrics have Markov type 2

Run `fa_banach_001`; source arXiv:math/0410422

### Source question

Naor–Peres–Schramm–Sheffield introduced and studied Markov type methods for smooth Banach spaces, trees, hyperbolic spaces, and several metric families. In the discussion section they ask, in belief form, whether planar graphs and doubling spaces have Markov type 2. The same item also mentions CAT(0) spaces; this packet does not address that adjacent part.

7. As discussed in Section 7 we believe that planar graphs and doubling spaces have Markov type 2. Also, it seems likely that CAT(0) spaces have Markov type 2 (see 8 for a discussion of CAT(0) spaces).

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Figure 1: arXiv:math/0410422, page 22, item 7 in the discussion section.

The target status question recorded here is:

Do planar graph metrics and doubling metric spaces have Markov type 2?

### Supporting literature status

Ding–Lee–Peres later introduced threshold embeddings as a tool for controlling Markov type. Their abstract states that if a metric space threshold-embeds into Hilbert space, then it has Markov type 2, and then draws the explicit consequence for planar graph metrics and doubling metrics. The abstract also says that this answers questions of Naor–Peres–Schramm–Sheffield.

The introduction of the supporting paper further identifies the exact NPSS beliefs: it says that Naor–Peres–Schramm–Sheffield had stated that planar graph metrics and doubling metrics should have Markov type 2, while only Markov type  $2 - \varepsilon$  was known there, and then says the paper resolves these questions.

### Idea of the proof identification

This packet is not an original proof. The identification is direct: the source paper asks for Markov type 2 of planar graph metrics and doubling spaces, and the supporting paper explicitly proves those two assertions and says they answer questions of Naor–Peres–Schramm–Sheffield.

At a high level, the supporting proof works by relaxing bi-Lipschitz embeddings to threshold embeddings. Ding–Lee–Peres prove that threshold embedding into Hilbert space is enough to force Markov type 2. They then use known random-partition and threshold-embedding results for planar metrics and doubling metrics to obtain the desired Markov type 2 conclusion.

**Theorem 1** (Literature answer). *Planar graph metrics and doubling metric spaces have Markov type 2.*

# Markov type and threshold embeddings

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## Abstract

For two metric spaces  $X$  and  $Y$ , say that  $X$  *threshold-embeds into*  $Y$  if there exist a number  $K > 0$  and a family of Lipschitz maps  $\{\varphi_\tau : X \rightarrow Y : \tau > 0\}$  such that for every  $x, y \in X$ ,

$$d_X(x, y) \geq \tau \implies d_Y(\varphi_\tau(x), \varphi_\tau(y)) \geq \|\varphi_\tau\|_{\text{Lip}} \tau / K,$$

where  $\|\varphi_\tau\|_{\text{Lip}}$  denotes the Lipschitz constant of  $\varphi_\tau$ . We show that if a metric space  $X$  threshold-embeds into a Hilbert space, then  $X$  has Markov type 2. As a consequence, planar graph metrics and doubling metrics have Markov type 2, answering questions of Naor, Peres, Schramm, and Sheffield. More generally, if a metric space  $X$  threshold-embeds into a  $p$ -uniformly smooth Banach space, then  $X$  has Markov type  $p$ . Our results suggest some non-linear analogs of Kwapien’s theorem. For instance, a subset  $X \subseteq L_1$  threshold-embeds into Hilbert space if and only if  $X$  has Markov type 2.

Figure 2: arXiv:1208.6088, page 1: the abstract states the planar and doubling Markov type 2 conclusion and identifies it as answering Naor–Peres–Schramm–Sheffield.

*Proof.* This is the explicit consequence stated in the abstract of Ding–Lee–Peres [2]. The same paper’s introduction identifies the claim as resolving the corresponding questions from Naor–Peres–Schramm–Sheffield [1]. Hence the planar and doubling parts of the source discussion item are answered affirmatively in later literature.  $\square$

## Verification notes and limitations

The source paper and supporting paper are distinct. The supporting paper explicitly names Naor–Peres–Schramm–Sheffield when saying that planar graph metrics and doubling metrics have Markov type 2, and its introduction identifies the same questions from the source paper.

This packet does not claim any status for the CAT(0) part of the source item. It also does not address the other discussion questions in arXiv:math/0410422, including the nonlinear Maurey constant, the  $L_1$ -valued extension question, the best tree constant, or maximal Markov type.

Bounded duplicate/status check: I searched the run’s cheap indexes for arXiv id 0410422, the source title, Markov type, planar graphs, doubling spaces, and nearby title keywords, finding no prior packet or attempt for this exact target. I searched local parsed sources and references for 1208.6088, “Markov type and threshold embeddings”, and “Ding Lee Peres”, locating the supporting paper and publication data. A web search over the exact supporting-paper title and target phrases found the same answer source and no conflicting status for the planar/doubling result.

This packet is a literature-status record only. It should not be counted as a new proof, partial result, or counterexample produced by this run.

## References

- [1] A. Naor, Y. Peres, O. Schramm, and S. Sheffield, *Markov chains in smooth Banach spaces and Gromov hyperbolic metric spaces*, arXiv:math/0410422, 2004.
- [2] J. Ding, J. R. Lee, and Y. Peres, *Markov type and threshold embeddings*, Geom. Funct. Anal. 23(4), 1207–1229, 2013; arXiv:1208.6088.